Christopher Sterbenz

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LLM TRAINING PLATFORM | LLM INFERENCE PLATFORM | MLOPS
AI SOFTWARE ENGINEER

AI infrastructure and platform engineer building distributed systems for GPU computing, multi-cloud inference, and frontier AI research. Focused on reliable, efficient systems that accelerate AI development and improve its impact on society.

SKILLS

Distributed Control Plane:	Multi-cluster routing · fleet load balancing · unified APIs · resource lifecycles
Kubernetes Architecture:	Kubernetes · operators/controllers · CRDs · reconciliation · custom autoscalers
GPU Infrastructure:	Slurm · SUNK · Run:ai · GPU scheduling · capacity planning
Agent Platforms:	Agent harnesses · sandboxed runtimes · MCP · session lifecycle · agent fleets
Cloud Infrastructure:	AWS · EKS · CoreWeave · multi-cloud architecture
Infrastructure & Delivery:	Terraform · Terragrunt · Helm · Argo CD · GitHub Actions
Backend Engineering:	FastAPI · REST APIs · PostgreSQL · asynchronous services · caching
Messaging & Async Tasks:	Redis · Celery · distributed workers · job failure recovery
Platform Security:	Workload identity · WIF · IAM/RBAC · NetworkPolicy · audit trails
Observability:	Prometheus · Grafana · Loki · PromQL · NVIDIA DCGM
Languages:	Python · Rust · C++ · Go · Bash

PROFESSIONAL EXPERIENCE

Staff AI Infrastructure Engineer, Biohub (Chan Zuckerberg Initiative) — 2026–Present

- Pioneered the agent platform adopted company-wide for both interactive and autonomous sessions
  - Drove adoption for the platform across leadership, researcher, and engineering stakeholders
  - Ensured agent activity is attributable and controllable: workload identity, scoped access controls, audit trails
  - Delivered remote sessions with automatic provisioning and lifecycle management for research workflows
  - Democratized cluster use for non-technical research parties by providing seamless remote chat interaction
  - Reduced vendor lock-in with model-provider-agnostic harnesses
  - Addressed ballooning model costs with an intelligent model selection router layer
- Owned strategic direction of multi-cluster, heterogeneous training platform: data movement and job routing
  - Achieved sub-hour training platform spin up times on arbitrary neocloud clusters: ArgoCD, Helm, Terraform
  - Scaled training infra to 6,000+ H100 equivalents, while designing to handle an additional 5x growth
  - Managed dynamic data transfer tooling, optimizing for S3 to cluster and inter-cluster use cases
  - Designed a unified API for scheduling training workloads across the cluster fleet

Senior AI Infrastructure Engineer, Biohub (Chan Zuckerberg Initiative) — 2024-2025

- Built a multi-cloud inference platform spanning AWS, CoreWeave, and Databricks
  - Reduced pre-training evaluation duration by 50x on L40 GPUs
  - Unified model deployment and serving through Python APIs and cluster-local execution services
  - Developed Kubernetes orchestration for GPU allocation, replica scaling, and workload placement
  - Automated model deployment and versioning with MLflow, Helm, and GitHub Actions
  - Implemented distributed batch inference with Redis and Celery, including real-time progress tracking
  - Integrated S3 and MinIO for inference input data and output artifacts.
  - Instrumented inference services with Prometheus metrics, endpoint health monitoring, and correlation IDs
- Owned design and engineering of researcher-friendly interface for NVIDIA RunAI CoreWeave cluster
  - Led team of 4 engineers including 2 mentorship roles to deliver total rework of cluster tooling on time
  - Balanced needs and wishlists of hundreds of researchers across dozens of teams
  - Abstracted cluster-facing operations to API layer, allowing for significantly higher engineering velocity
  - Enabled on-call traceability by adding robust debug info to user CLI commands on Loki backend
  - Increased researcher quality of life via CLI error auto-healing and automatic debugging suggestions
  - Stabilized RunAI workspace connection with a custom jump-host implementation

Machine Learning Engineer, Twosense, Inc. — 2021–2024

- Achieved 85% reduction of model training costs while accommodating a 1,000x increase in data volume
- Owned the transition of large, GPU-based training jobs to Kubernetes, Kubeflow hosted on AWS EKS
- Reduced model training memory footprint by 100x by introducing more efficient data loading
- Strengthened production reliability with infrastructure and data anomaly alerting via Grafana and Slack
- Migrated Cloud infrastructure setup to Terraform to achieve high availability and drift detection
- Collaborated closely with researchers to rapidly ship novel model architectures
- Initiated a continuous delivery system to tie together research, development, and production tasks
- Enhanced customer privacy by eliminating the need for sensitive data in the feature store
- Formulated a standard, statistically driven template for research tasks to ensure transferability of results

Machine Learning Engineer, SciTec, Inc. — 2018–2021

- Extended Super-Resolution architecture to a recurrent approach based on ingesting satellite video feeds
- Optimized Super-Resolution training to achieve convergence using 25% of the original time and resources
- Implemented a Keras LSTM Deep Learning alternative to SciTec's Geospatial threat classifier
- Led regular company-wide workshops on various Deep Learning and Artificial Intelligence topics
- Proposed new business solutions using Deep Multi-Agent Reinforcement Learning to pilot C-UAVs
- Deployed new technology to multiple disparate environments using Docker containerization
- Maintained real-time C++ classification architecture as lead developer for two years
- Extracted highly coupled legacy software into the software as a service (SaaS) paradigm
- Coordinated team of five developers using Agile project management tools such as Jira and GitLab
- Built Continuous Integration Pipeline and achieved a persistent 95% code coverage with Google Test

Mobile Website Development Contractor, Blue Hill Data Services; Pearl River, NY — Summer 2017

- Developed a dynamic website to provide clients with a seamless experience on all platforms
- Worked with top-level executives to ensure design accuracy to specifications

IT Support, ResNet TCNJ; Ewing, NJ — 2016-2018

- Acted as a customer-facing specialist to diagnose and address a variety of technical problems

FORMAL EDUCATION

Bachelor of Science (BS) Computer Science, The College of New Jersey, Ewing, NJ — 2014–2018
Statistics Minor